

└ s - input size, k - kernel size, p - padding, t - stride

└ $p = k // 2$, k is odd, s is divisible by t (hw 1 requirements)

than: $S_{out} = \left\lfloor \frac{s + 2p - k}{t} \right\rfloor + 1 = \left\lfloor \frac{s - 1}{t} \right\rfloor + 1 = \frac{s}{t}$ - out size

~~per~~ for conv and pool layers.

└ 1 MAC = 2 FLOPs, 1 comparison = 1 FLOP, c_{in} - input channels,

c_{out} - output channels, B - batch size, than:

$$FLOPs_{conv} = \underbrace{S_{out}^2 \cdot C_{in}}_{\text{apply 1 kernel to all steps and channels}} \cdot \underbrace{2k^2}_{\substack{k^2 \text{ MAC} \\ \text{per each kernel} \\ \text{application}}} \cdot \underbrace{C_{out}}_{\text{repeat per each kernel}} \cdot \underbrace{B}_{\text{per each item}}$$

$$FLOPs_{maxpool} = S_{out}^2 \cdot C_{in} \cdot \underbrace{(k^2 - 1)}_{\substack{k^2 - 1 \text{ comparisons} \\ \text{per window}}} \cdot B$$

$$FLOPs_{avgpool} = \underbrace{S^2}_{(S^2 - 1) \text{ adds and 1 divide in each channel}} \cdot C_{in} \cdot B$$

$$FLOPs_{relu} = \underbrace{S^2 \cdot C_{in}}_{1 \text{ comp. per value}} \cdot B$$

$$FLOPs_{linear} = \underbrace{2 \cdot C_{in}}_{C_{in} \text{ mults, } (C_{in} - 1) \text{ adds and 1 bias add}} \cdot C_{out} \cdot B$$

Flatten uses view, no FLOPs.

$FLOPs(S, B)$ than just sum over all layers with given $\rightarrow$

→ $k^i, p^i, t^i, c_{in}^i, c_{out}^i, s_{in}^i, s_{out}^i$, where i - is layer number, $s_{in/out}^i$ depends on s : $FLOPs(s, B) = \sum_i FLOPs_i$

Memory allocated includes:

- 1) weights
- 2) input batch x
- 3) cuBLAS workspace
- 4) input and output of current layer

$$W_{conv} = \underbrace{C_{in} \cdot k^2}_{\text{per kernel}} \cdot C_{out} \cdot \underbrace{4}_{\text{fp32}}$$

$$W_{linear} = (C_{in} \cdot C_{out} + \underbrace{C_{out}}_{\text{bias}}) \cdot 4$$

$$x = 3s^2 \cdot B \cdot 4$$

$$W_{cuBLAS} = 4096 \cdot 1024 \cdot 2 + 16 \cdot 1024 \cdot 8 \leftarrow \text{default for sm75 (T4 colab)}$$

$$A_{conv} = (C_{in} s^2 + C_{out} s_{out}^2) \cdot B \cdot 4$$

$$A_{maxpool} = (4s^2 + 4s_{out}^2 + \underbrace{8s_{out}^2}_{\substack{\text{indices in} \\ \text{max-pooled_with_indices}}}) \cdot C_{in} \cdot B$$

$$A_{avgpool} = (s^2 + 1) \cdot C_{in} \cdot B \cdot 4$$

$$A_{~~relu~~ relu} = s^2 \cdot C_{in} \cdot B \cdot 4$$

$$A_{linear} = (C_{in} + C_{out}) \cdot B \cdot 4$$

$$\text{Memory}(S, B) = \sum_i W_i + W_{\text{cuBLAS}} + \max(A_1, x + \max_{j \geq 2} A_j)$$

Each kernel reads input and weight element from DRAM once and writes output element once, then:

$$M_{\text{conv}} = \left(\underbrace{(C_{\text{in}} S^2)}_{\text{input}} + \underbrace{(C_{\text{out}} \cdot S_{\text{out}}^2)}_{\text{output}} \right) \cdot B + \underbrace{(C_{\text{in}} k^2 C_{\text{out}})}_{\text{weights}} \cdot 4$$

$$M_{\text{linear}} = \left((C_{\text{in}} + C_{\text{out}}) \cdot B + C_{\text{in}} \cdot C_{\text{out}} + C_{\text{out}} \right) \cdot 4$$

$$M_{\text{relu}} = 2S^2 \cdot C_{\text{in}} \cdot B \cdot 4$$

$$M_{\text{maxpool}} = A_{\text{maxpool}}$$

$$M_{\text{avgpool}} = A_{\text{avgpool}}$$

$$\text{Bytes}(S, B) = \sum_i M_i$$

Each layer except Flatten launches 1 CUDA kernel. Each kernel takes some time t_{launch} to launch, let's assume it's same for all kernels, then: $\text{Latency}(S, B, \Theta) = \sum_i (t_{\text{launch}} + \max(F_i/P, M_i/w))$ where P - FLOP/s, w - DRAM bytes/s. $\Theta = (t_{\text{launch}}, P, w)$

$$\text{Energy}(S, B, \Theta_E) = e_f \cdot \text{FLOPs} + e_m \cdot \text{Bytes} + P_{\text{static}} \cdot \text{Latency}$$

where: e_f - J/FLOP, e_m - J/byte, P_{static} - J/s.

$$\Theta_E = (e_f, e_m, P_{\text{static}}).$$